

**Laboratorium**  
**Multimedia dan Internet of Things**  
**Departemen Teknik Komputer**  
*Institut Teknologi Sepuluh Nopember*

# **Laporan Akhir**

## **Praktikum Jaringan Komputer**

**VLAN, Trunk, OSPF, MultiVendor**

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2026

# 1 Tahapan Praktikum

## Praktikum 1: Konfigurasi Cisco Switch

### Langkah 1: Masuk ke terminal Cisco Switch

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

### Langkah 2: Membuat VLAN

```
vlan 10
  name Finance
exit

vlan 20
  name HR
exit

vlan 30
  name IT
exit

vlan 40
  name Marketingg
exit
```

Verifikasi menggunakan perintah berikut untuk memastikan vlan sudah dibuat dan statusnya aktif:

```
show vlan brief
```



| VLAN | Name               | Status    | Ports                                                    |
|------|--------------------|-----------|----------------------------------------------------------|
| 1    | default            | active    | Gi0/0, Gi0/1, Gi0/2, Gi0/3<br>Gi1/0, Gi1/1, Gi1/2, Gi1/3 |
| 10   | Finance            | active    |                                                          |
| 20   | HR                 | active    |                                                          |
| 30   | IT                 | active    |                                                          |
| 40   | Marketing          | active    |                                                          |
| 1002 | fddi-default       | act/unsup |                                                          |
| 1003 | token-ring-default | act/unsup |                                                          |
| 1004 | fddinet-default    | act/unsup |                                                          |
| 1005 | trnet-default      | act/unsup |                                                          |

```
Switch#
```

### Langkah 3: Konfigurasi Access Port untuk Setiap VLAN

Port Access harus disesuaikan dengan interface pada cisco switch.

```
Switch#conf t

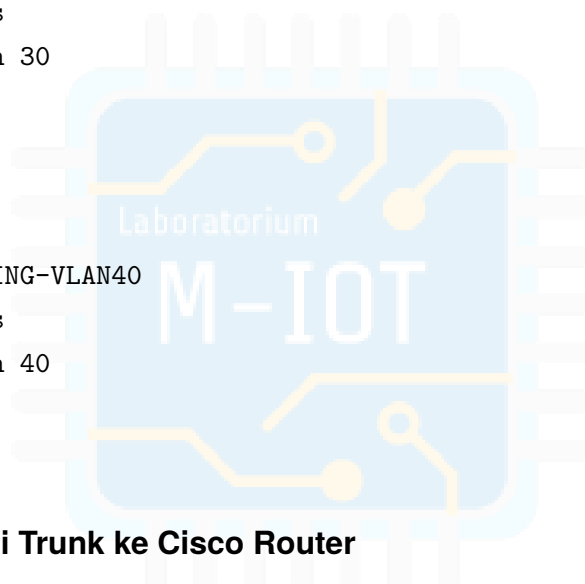
interface gi0/2
```

```
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
exit
```

```
interface gi0/1
description PC-HR-VLAN20
switchport mode access
switchport access vlan 20
no shutdown
exit
```

```
interface gi1/1
description PC-IT-VLAN30
switchport mode access
switchport access vlan 30
no shutdown
exit
```

```
interface gi0/3
description PC-MARKETING-VLAN40
switchport mode access
switchport access vlan 40
no shutdown
exit
```



#### **Langkah 4: Konfigurasi Trunk ke Cisco Router**

Konfigurasi trunk port pada interface Gi0/0 dari switch menuju Cisco Router Gi0/2 untuk membawa vlan 10, 20, 30, dan 40.

```
Switch#conf t
```

```
interface gi0/0
switchport trunk encapsulation dot1q
description TRUNK-TO-CISCO-ROUTER
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
exit
```

Verifikasi hasil trunk port menggunakan perintah berikut:

```
show interfaces trunk
```

| Port  | Mode | Encapsulation | Status   | Native vlan |
|-------|------|---------------|----------|-------------|
| Gi0/0 | on   | 802.1q        | trunking | 1           |

| Port  | Vlans allowed on trunk |
|-------|------------------------|
| Gi0/0 | 10,20,30,40            |

| Port  | Vlans allowed and active in management domain |
|-------|-----------------------------------------------|
| Gi0/0 | 10,20,30,40                                   |

| Port  | Vlans in spanning tree forwarding state and not pruned |
|-------|--------------------------------------------------------|
| Gi0/0 | 10,20,30,40                                            |

Save semua konfigurasi dengan perintah:

```
write memory
```

## Praktikum 2 Konfigurasi Cisco Router

### 1. Masuk ke Cisco Router

```
enable
configure terminal
```

### 2. Konfigurasi Interface Trunk ke Switch

Pilih interface yang terhubung ke Cisco Switch. Pada topologi, interface yang terhubung is Gig0/2

```
interface gig0/2
description TRUNK-TO-CISCO-SWITCH
no ip address
no shutdown
exit
```

### 3. Konfigurasi Subinterface VLAN 10

```
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

### 4. Konfigurasi Subinterface VLAN 20

```
interface gig0/2.20
description GATEWAY-VLAN20-HR
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

## 5. Konfigurasi Subinterface VLAN 30

```
interface gig0/2.30
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

## 6. Konfigurasi Subinterface VLAN 40

```
interface gig0/2.40
description GATEWAY-VLAN40-MARKETING
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```

## 7. Konfigurasi DHCP-Server untuk Vlan 10

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp pool VLAN10-FINANCE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

## 8. Konfigurasi DHCP-Server untuk Vlan 20

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

## 9. Konfigurasi IP Static pada VPCS untul vlan 30 dan 40

PC Vlan 30 :

```
ip 192.168.30.10/24 192.168.30.1
```

```

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

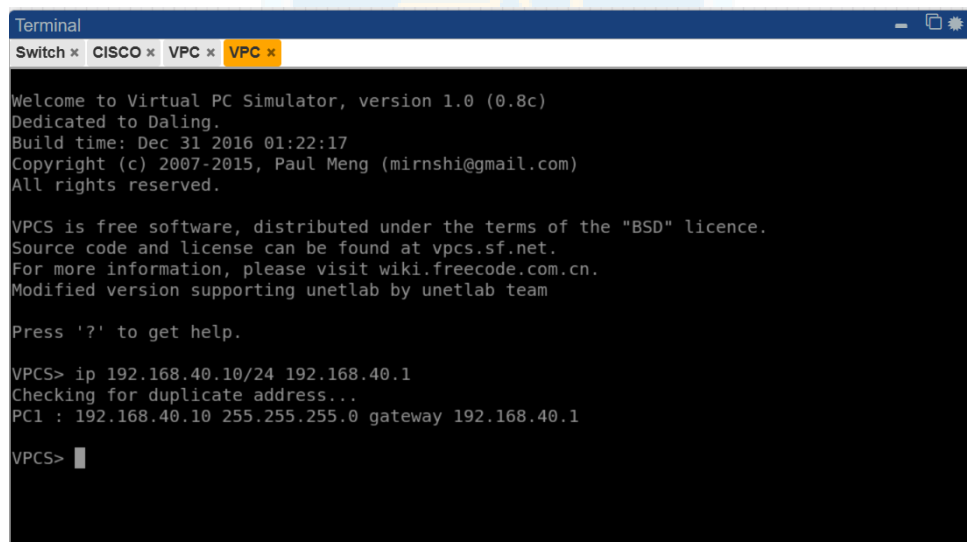
VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> █

```

PC Vlan 40 :

ip 192.168.40.10/24 192.168.40.1



```

Terminal
Switch x CISCO x VPC x VPC x

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> █

```

PC Vlan 10 dan 20 menggunakan dhcp:

ip dhcp

```
Terminal
Switch x CISCO x VPC x VPC x VPC x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
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VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS>
```

```
Terminal
Switch x CISCO x VPC x VPC x VPC x

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

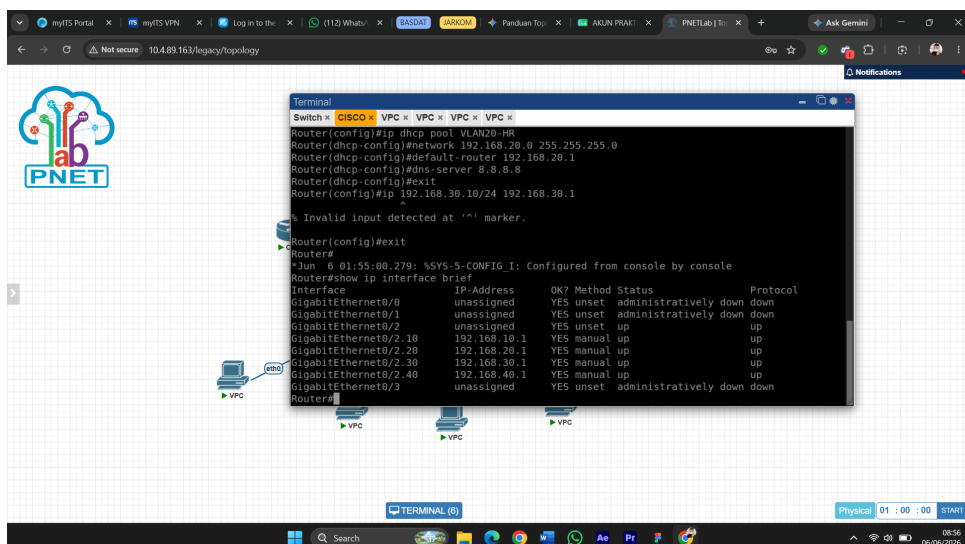
VPCS>
VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS>
```

## 10. Verifikasi Cisco Router

Cek interface:

show ip interface brief



The screenshot shows a PNET virtual network simulator interface. On the left, there is a PNET logo and a network diagram with several VPCs connected to a central router. The main window displays a terminal window with the following content:

```
Terminal
Switch x CISCO x VPC x VPC x VPC x

Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip 192.168.30.10/24 192.168.30.1
Router#

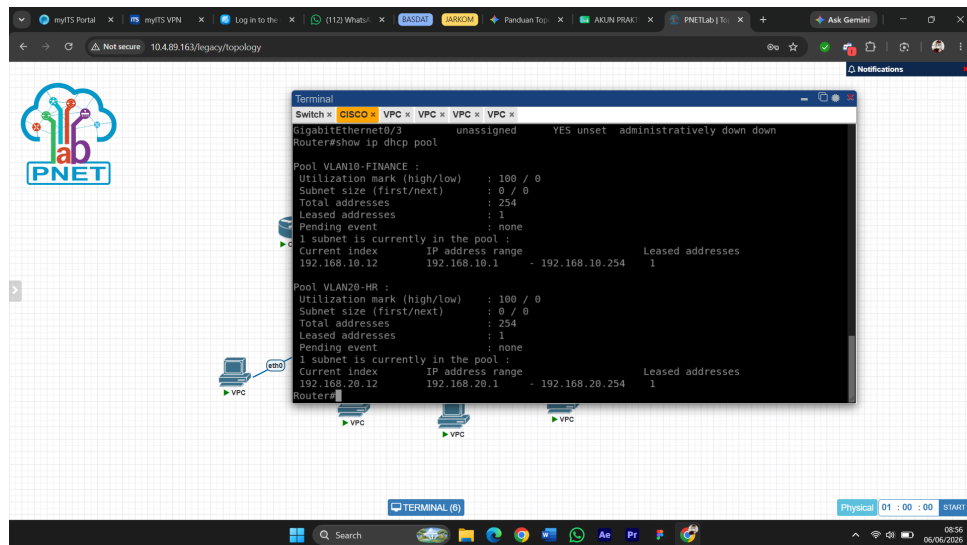
^
^ Invalid input detected at '^' marker.

Router(config)#exit
Router#
Jun 6 01:55:00.279: %SYS-5-CONFIG I: Configured from console by console
Router#show ip interface brief

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned     YES unset  administratively down  down
GigabitEthernet0/1  unassigned     YES unset  administratively down  down
GigabitEthernet0/2  unassigned     YES unset  up          up
GigabitEthernet0/2.10  192.168.10.1  YES manual up          up
GigabitEthernet0/2.20  192.168.20.1  YES manual up          up
GigabitEthernet0/2.30  192.168.30.1  YES manual up          up
GigabitEthernet0/2.40  192.168.40.1  YES manual up          up
GigabitEthernet0/3  unassigned     YES unset  administratively down  down
Router#
```

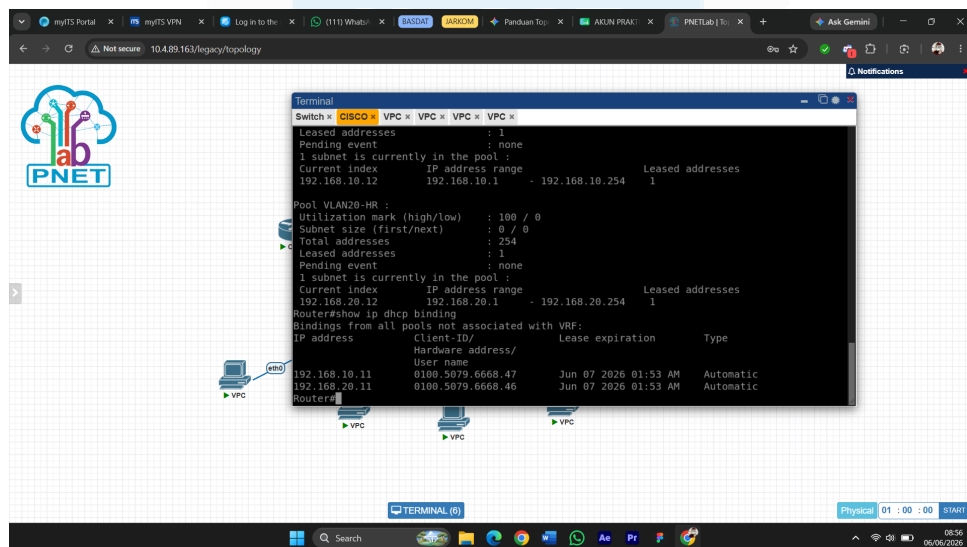
Cek dhcp pool:

show ip dhcp pool



Cek client DHCP yang mendapatkan IP:

show ip dhcp binding



## 11. Pengujian dari VPCS

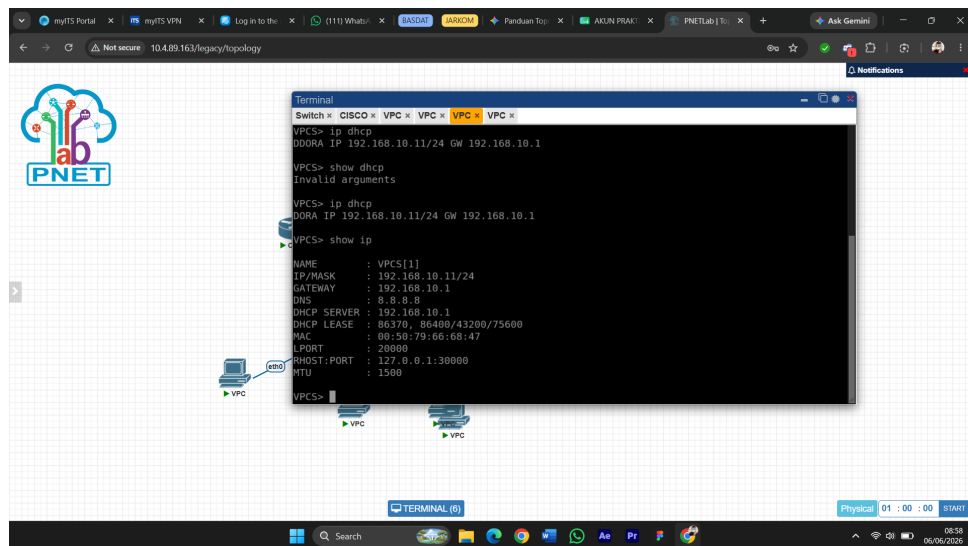
Masuk ke pc vlan 10 untuk request ip secara dhcp:

ip dhcp

Pastikan pc vlan 10 sudah dapat ip secara dhcp dengan command:

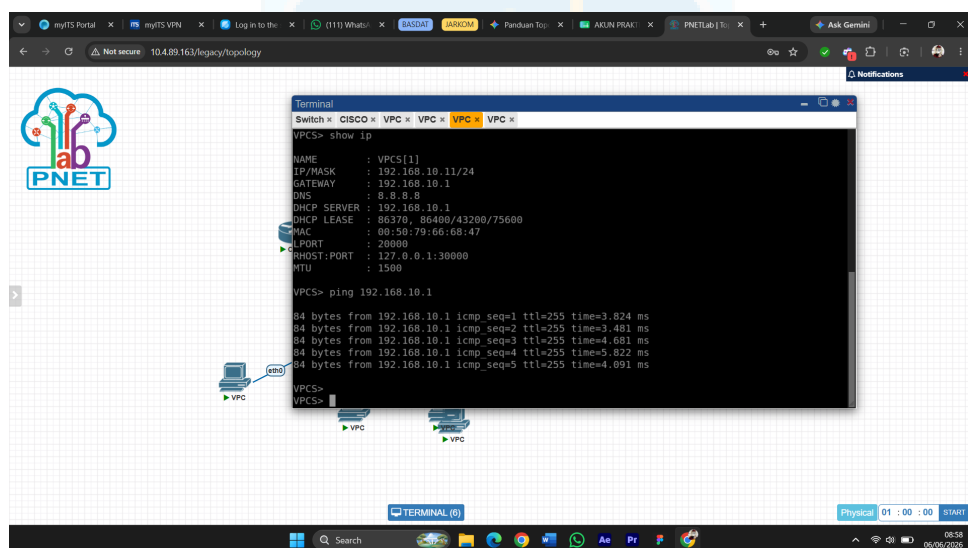
show ip





Dari pc vlan 10 ping gateway 192.168.10.1:

ping 192.168.10.1

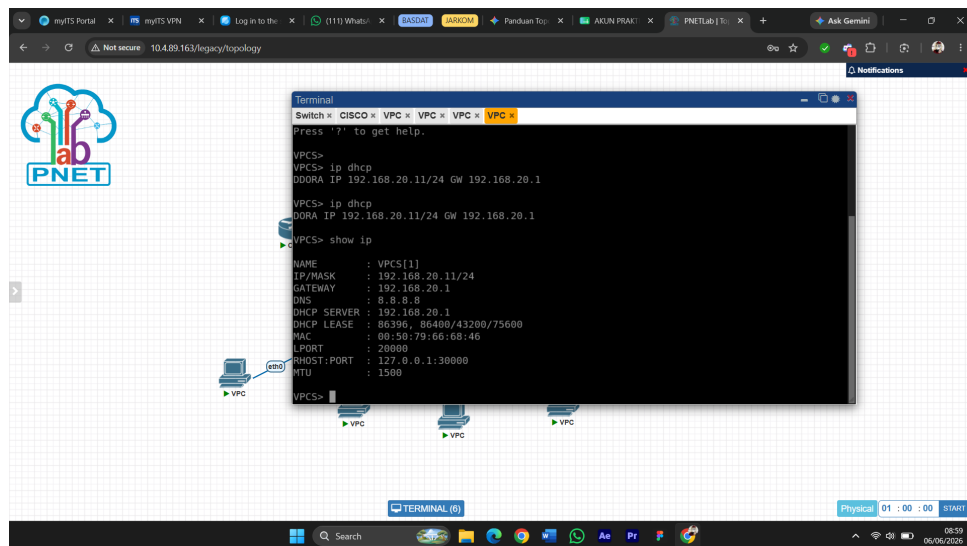


Masuk ke pc vlan 20 untuk request ip secara dhcp:

ip dhcp

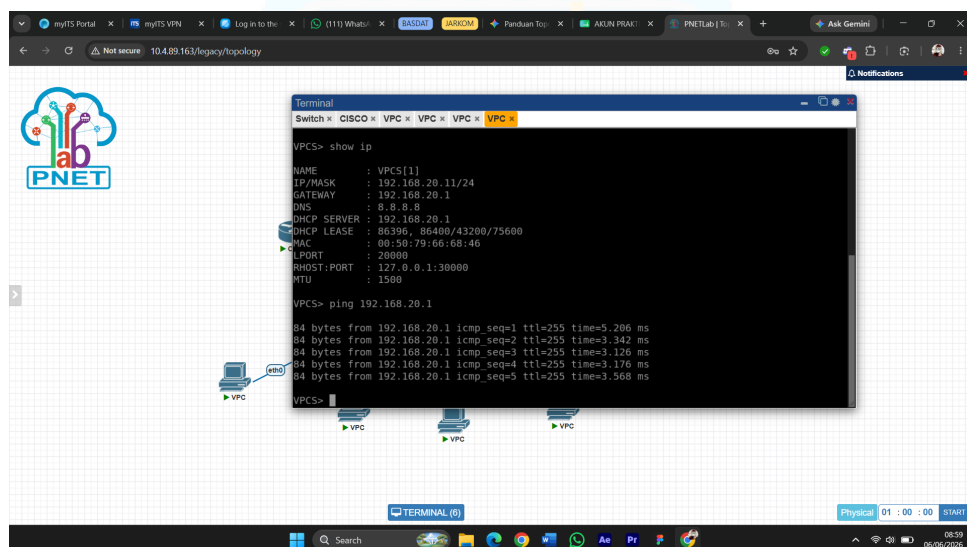
Pastikan pc vlan 20 sudah dapat ip secara dhcp with command:

show ip



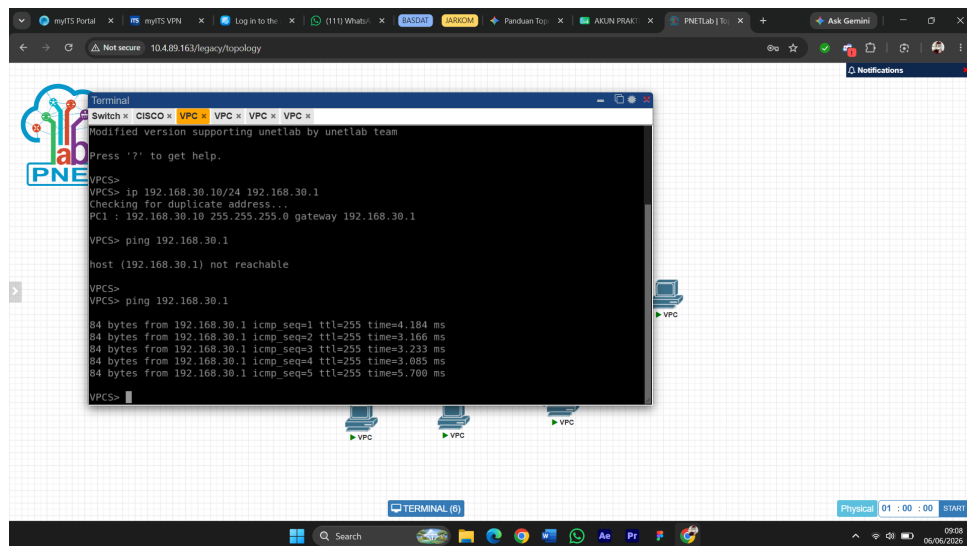
Dari pc vlan 20 ping gateway 192.168.20.1:

ping 192.168.20.1



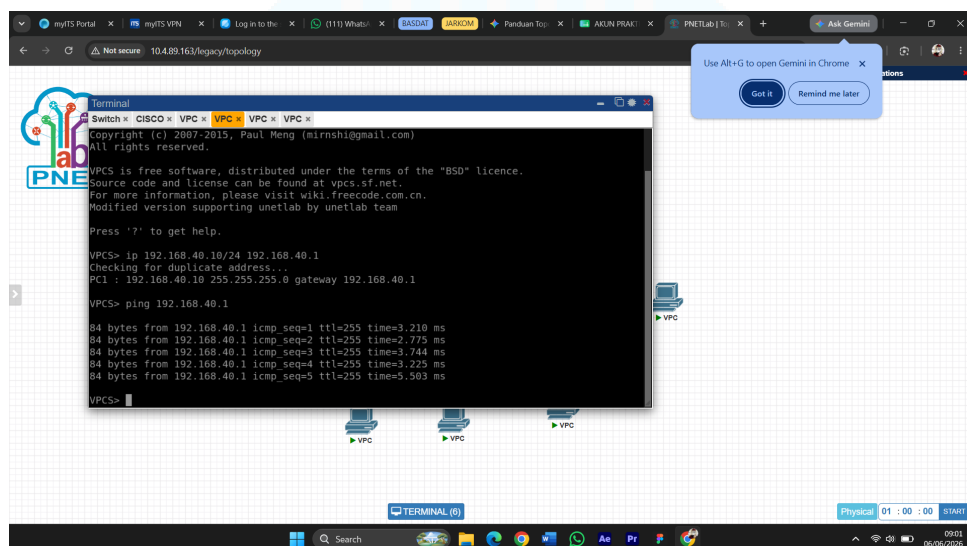
Dari pc vlan 30 ping gateway 192.168.30.1:

ping 192.168.30.1



Dari pc vlan 40 ping gateway 192.168.40.1:

ping 192.168.40.1

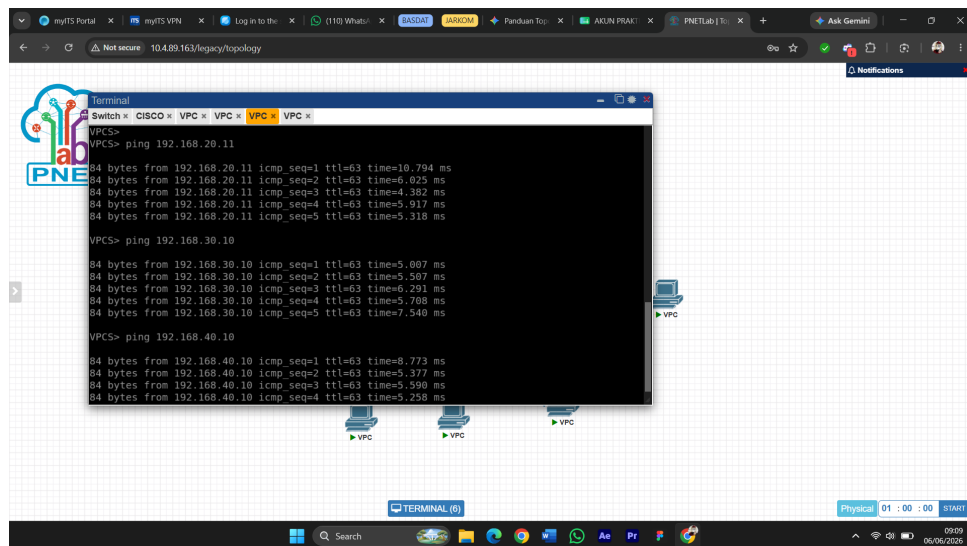


Uji ping antar-Vlan dari vlan 10:

ping 192.168.20.11

ping 192.168.30.10

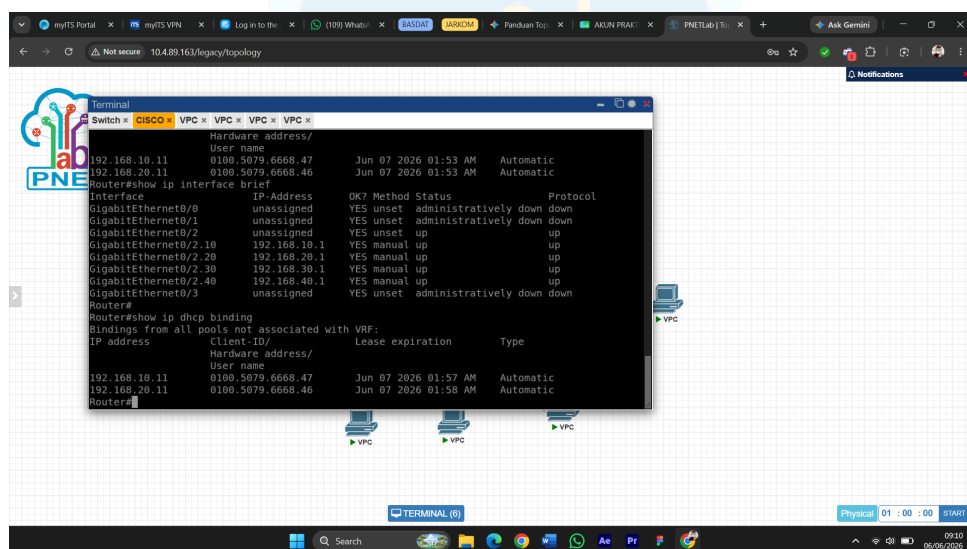
ping 192.168.40.10



Masuk ke cisco router lagi untuk melihat ip dhcp binding:

show ip dhcp binding

Amati IP yang diberikan oleh router ke client sesuai dengan ip di pc vlan 10 dan 20.



## Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

### 1. Konfigurasi Lan Huawei

konfigurasi ip address untuk lan huawei:

```
system-view
```

```
interface Ethernet1/0/2
```

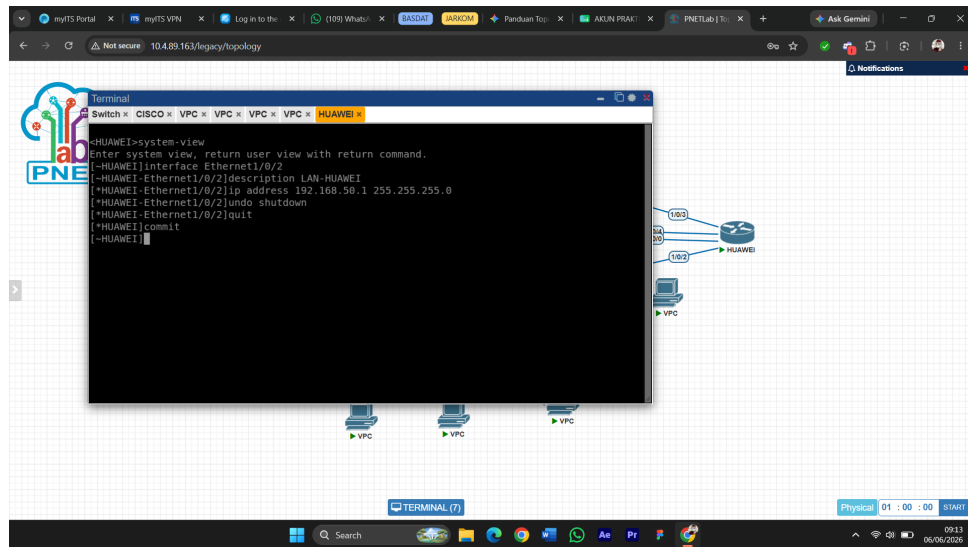
```
description LAN-HUAWEI
```

```
ip address 192.168.50.1 255.255.255.0
```

```
undo shutdown
```

```
quit
```

commit



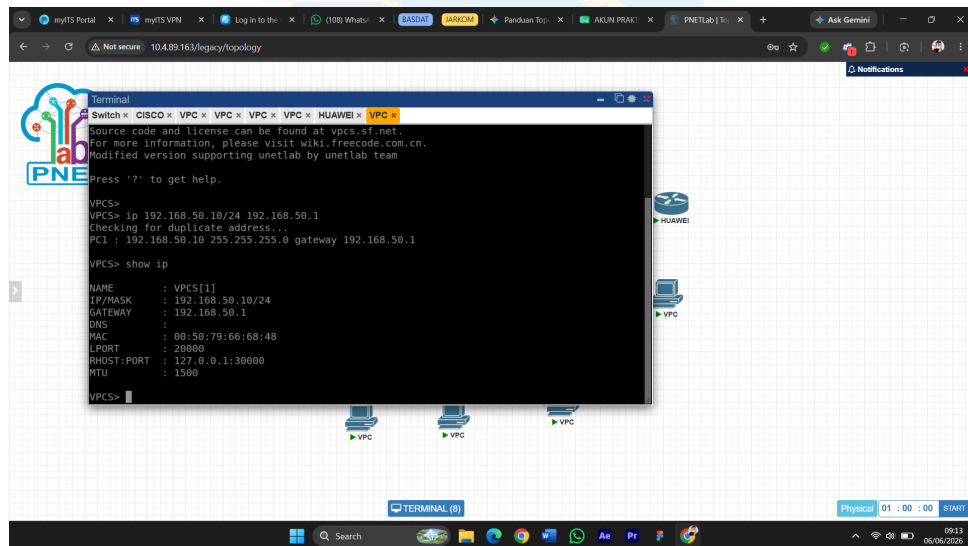
## 2. Konfigurasi IP VPC Lan Huawei

Pada pc lan huawei:

ip 192.168.50.10/24 192.168.50.1

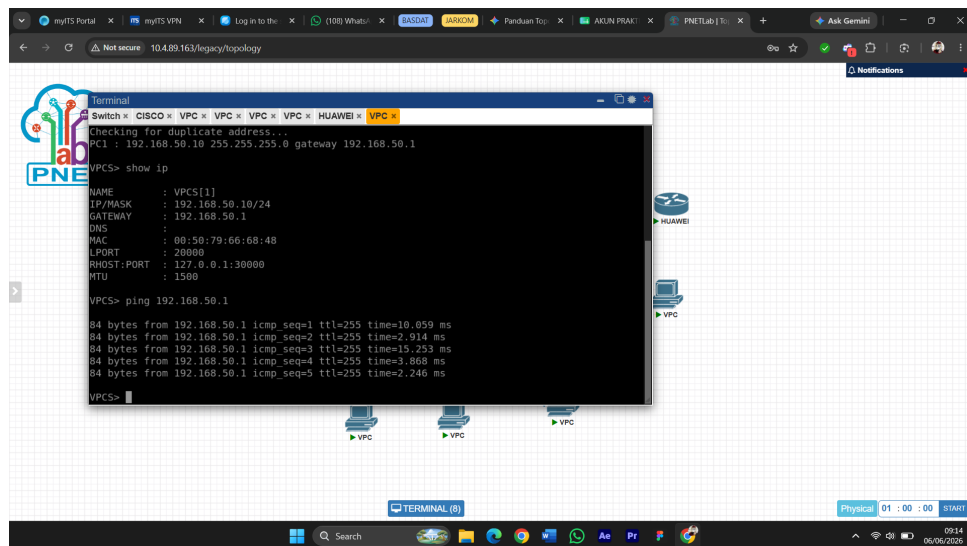
Verifikasi:

show ip



Testing ping gateway huawei:

ping 192.168.50.1



### 3. Konfigurasi IP link cisco <-> Huawei link 1

Network:

10.10.10.0/30

#### Cisco Router

enable

configure terminal

interface gi0/3

description LINK1-TO-HUAWEI

ip address 10.10.10.1 255.255.255.252

no shutdown

exit

#### Huawei Router

system-view

interface Ethernet1/0/0

description LINK1-TO-CISCO

ip address 10.10.10.2 255.255.255.252

undo shutdown

quit

commit

### 4. Konfigurasi IP Link Cisco Huawei Link 2

Network:

10.10.11.0/30

## Cisco Router

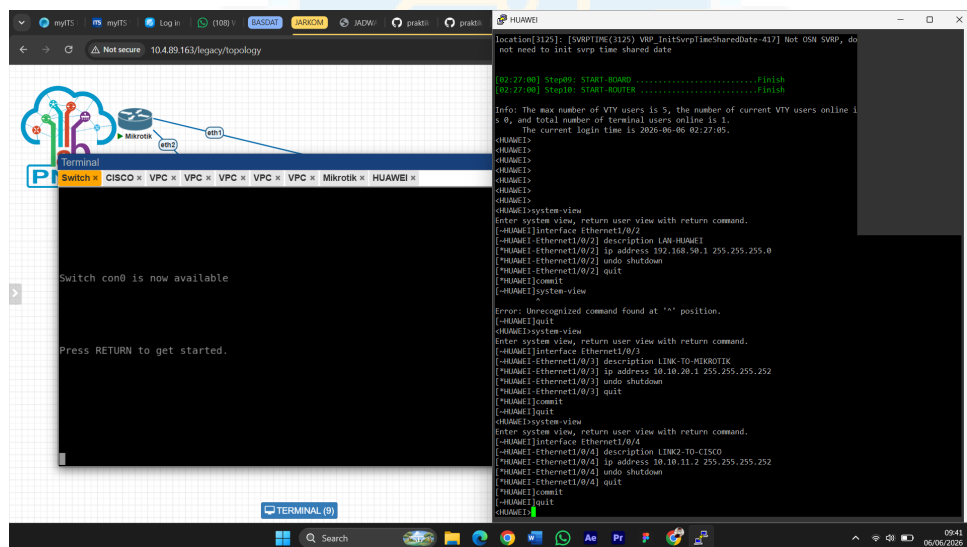
```
interface gi0/0
description LINK2-TO-HUAWEI
ip address 10.10.11.1 255.255.255.252
no shutdown
exit
```

## Huawei Router

```
system-view

interface Ethernet1/0/4
description LINK2-TO-CISCO
ip address 10.10.11.2 255.255.255.252
undo shutdown
quit

commit
```



## 5. Konfigurasi IP link Huawei <-> Mikrotik

Network:

10.10.20.0/30

## Huawei Router

```
system-view

interface Ethernet1/0/3
description LINK-TO-MIKROTIK
ip address 10.10.20.1 255.255.255.252
```

```
undo shutdown
quit

commit
```

```
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]commit
[~HUAWEI]system-view
      ^
Error: Unrecognized command found at '^' position.
[~HUAWEI]quit
<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3] description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3] undo shutdown
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]commit
```

## Mikrotik

```
/ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
```

## 6. Konfigurasi IP link Cisco <-> Mikrotik

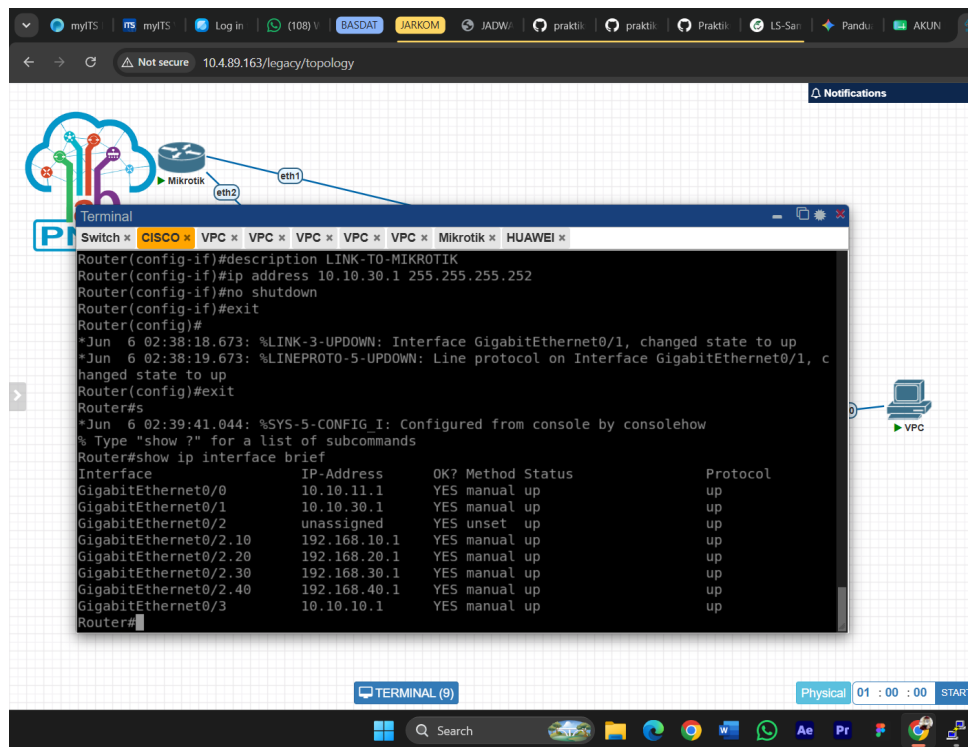
Network:

10.10.30.0/30

### Cisco Router

```
interface gi0/1
description LINK-TO-MIKROTIK
ip address 10.10.30.1 255.255.255.252
no shutdown
exit
```





## Mikrotik

/ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO

## 7. Verifikasi Interface

### Cisco Router

show ip interface brief

### Huawei Router

display ip interface brief

the number of interface that is down in Protocol is 1

| Interface          | IP Address/Mask  | Physical | Protocol | VPN   |
|--------------------|------------------|----------|----------|-------|
| Ethernet1/0/0      | 10.10.10.2/30    | up       | up       | --    |
| Ethernet1/0/0.4094 | 128.1.138.137/16 | up       | up       | 13vpn |
| Ethernet1/0/1      | unassigned       | up       | down     | --    |
| Ethernet1/0/1.4094 | 128.1.138.137/16 | up       | up       | 13vpn |
| Ethernet1/0/2      | 192.168.50.1/24  | up       | up       | --    |
| Ethernet1/0/2.4094 | 128.1.138.137/16 | up       | up       | 13vpn |
| Ethernet1/0/3      | 10.10.20.1/30    | up       | up       | --    |
| Ethernet1/0/3.4094 | 128.1.138.137/16 | up       | up       | 13vpn |
| Ethernet1/0/4      | 10.10.11.2/30    | up       | up       | --    |
| Ethernet1/0/4.4094 | 128.1.138.137/16 | up       | up       | 13vpn |

---- More ----

## Mikrotik

/ip address print

## 8. Verifikasi ping Antar Router

### Dari Cisco Router

```
ping 10.10.10.2
ping 10.10.11.2
ping 10.10.30.2
```

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 3/4/9 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
```

## Dari Router Huawei

```
ping 10.10.10.1
ping 10.10.11.1
ping 10.10.20.2
```



```
<HUAWEI>ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=6 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/3/6 ms

<HUAWEI>ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

<HUAWEI>ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/1 ms
```

## Dari Router Mikrotik

```
ping 10.10.20.1
ping 10.10.30.1
```

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                            56 255 4ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 1ms
4 10.10.20.1                            56 255 1ms
5 10.10.20.1                            56 255 1ms
6 10.10.20.1                            56 255 1ms
7 10.10.20.1                            56 255 1ms
8 10.10.20.1                            56 255 1ms
9 10.10.20.1                            56 255 1ms
10 10.10.20.1                           56 255 1ms
11 10.10.20.1                           56 255 1ms
12 10.10.20.1                           56 255 1ms
13 10.10.20.1                           56 255 1ms
14 10.10.20.1                           56 255 1ms
sent=15 received=15 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                            56 255 2ms
1 10.10.30.1                            56 255 1ms
2 10.10.30.1                            56 255 1ms
3 10.10.30.1                            56 255 2ms
4 10.10.30.1                            56 255 2ms
5 10.10.30.1                            56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
```

## Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

### 1. Konsep routing ospf pada topologi

### 2. Network yang diiklankan ke ospf

#### Cisco Router

10.10.10.0/30  
10.10.11.0/30  
10.10.30.0/30  
192.168.10.0/24  
192.168.20.0/24  
192.168.30.0/24  
192.168.40.0/24

#### Huawei Router

10.10.10.0/30  
10.10.11.0/30  
10.10.20.0/30  
192.168.50.0/24

#### Mikrotik

10.10.20.0/30  
10.10.30.0/30

### 3. Konfigurasi ospf pada Cisco Router

Masuk ke Cisco Router:

```
enable
configure terminal
```

Konfigurasi OSPF:

```
router ospf 1
router-id 1.1.1.1
network 10.10.10.0 0.0.0.3 area 0
network 10.10.11.0 0.0.0.3 area 0
network 10.10.30.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
exit
```

Atur OSPF cost pada interface Cisco.

```
interface gi0/3
description LINK1-T0-HUAWEI
ip ospf cost 10
exit

interface gi0/0
description LINK2-T0-HUAWEI
ip ospf cost 10
exit

interface gi0/1
description LINK-T0-MIKROTIK
ip ospf cost 100
exit
```

Verifikasi OSPF Cisco:

```
show ip ospf neighbor
show ip route ospf
```

Hasil kemungkinan kosong karena router tetangga masih belum dikonfigurasi

```
show ip protocols
```

#### 4. Konfigurasi OSPF pada Huawei Router

Masuk ke huawei router:

```
system-view
```

Konfigurasi OSPF:

```
ospf 1 router-id 2.2.2.2
area 0.0.0.0
network 10.10.10.0 0.0.0.3
network 10.10.11.0 0.0.0.3
network 10.10.20.0 0.0.0.3
network 192.168.50.0 0.0.0.255
quit
quit
```

Atur OSPF cost pada interface huawei.

```
interface Ethernet1/0/0
description LINK1-T0-CISCO
ospf cost 10
quit
```

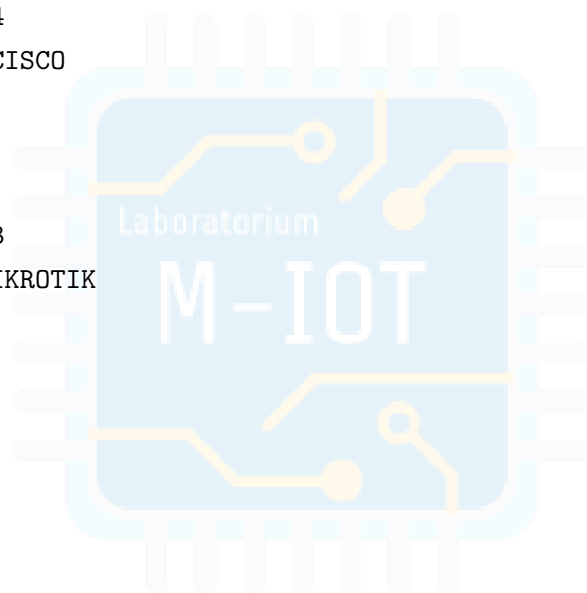
```
interface Ethernet1/0/4
description LINK2-T0-CISCO
ospf cost 10
quit
```

```
interface Ethernet1/0/3
description LINK-T0-MIKROTIK
ospf cost 100
quit
```

```
commit
```

Verifikasi OSPF Huawei:

```
display ospf peer
```



```
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave    Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2      MTU: 1500
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m06s
  Neighbor Up Time : 2026-06-06 02:55:12
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: Full                Mode:Nbr is Slave    Priority: 1
  DR: 10.10.11.1            BDR: None           MTU: 1500
  Dead timer due in 35 sec
  Retrans timer interval: 5
```

display ip routing-table protocol ospf

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
-----
10.10.30.0/30       OSPF   10   110     D   10.10.11.1   Ethernet1/0/4
                   OSPF   10   110     D   10.10.10.1   Ethernet1/0/0
192.168.10.0/24      OSPF   10   11       D   10.10.11.1   Ethernet1/0/4
                   OSPF   10   11       D   10.10.10.1   Ethernet1/0/0
192.168.20.0/24      OSPF   10   11       D   10.10.11.1   Ethernet1/0/4
                   OSPF   10   11       D   10.10.10.1   Ethernet1/0/0
192.168.30.0/24      OSPF   10   11       D   10.10.11.1   Ethernet1/0/4
                   OSPF   10   11       D   10.10.10.1   Ethernet1/0/0
192.168.40.0/24      OSPF   10   11       D   10.10.11.1   Ethernet1/0/4
                   OSPF   10   11       D   10.10.10.1   Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4
```

display ospf routing

```
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
---- More ----
```

## 5. Konfigurasi OSPF pada MikroTik

Pada topologi ini, Mikrotik digunakan sebagai jalur backup antara cisco dan huawei.

Masuk ke terminal Mikrotik, lalu konfigurasi router-id OSPF:

```
/routing ospf instance set default router-id=3.3.3.3
```

Tambahkan network OSPF:

```
/routing ospf network add network=10.10.20.0/30 area=backbone
/routing ospf network add network=10.10.30.0/30 area=backbone
```

Atur cost interface mikrotik menjadi jalur backup:

```
/routing ospf interface add interface=ether1 cost=100
/routing ospf interface add interface=ether2 cost=100
```

### Verifikasi Mikrotik

```
/routing ospf neighbor print
```

Cek routing table OSPF:

```
/ip route print where ospf
```

## 6. Verifikasi OSPF Neighbor

### Cisco Router

```
show ip ospf neighbor
```

```
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1     FULL/BDR        00:00:30    10.10.30.2     GigabitEthernet0/1
2.2.2.2          1     FULL/BDR        00:00:30    10.10.11.2     GigabitEthernet0/0
2.2.2.2          1     FULL/BDR        00:00:30    10.10.10.2     GigabitEthernet0/3
```

### Huawei Router

display ospf peer

```
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1          Address: 10.10.10.1
  State: Full              Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1          BDR: 10.10.10.2        MTU: 1500
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m32s
  Neighbor Up Time : 2026-06-06 02:55:12
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3          Address: 10.10.20.2
  State: Full              Mode:Nbr is Master     Priority: 1
  DR: 10.10.20.1          BDR: 10.10.20.2        MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
```

## Mikrotik

/routing ospf neighbor print

```
[admin@tikrofik] > /routing ospf instance set default router-id=3.3.3.3
[admin@tikrofik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@tikrofik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@tikrofik] > /routing ospf interface add interface=ether1 cost=100
[admin@tikrofik] > /routing ospf interface add interface=ether2 cost=100
[admin@tikrofik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="ExStart" state-changes=3
  ls-retransmits=0 ls-requests=0 db-summaries=0

1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=5s
[admin@tikrofik] >
```

## 7. Verifikasi Routing Table

**Cisco Router** Cek route menuju lan huawei:

```
show ip route 192.168.50.0
```

Cek semua route ospf:

```
show ip route ospf
```

**Huawei Router**

Cek semua route ospf:

```
display-ip routing table protocol ospf
```



## Mikrotik

/ip route print where ospf

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.20.1 110
    10.10.30.1
1 ADo 10.10.11.0/30      10.10.20.1 110
    10.10.30.1
2 ADo 192.168.10.0/24     10.10.30.1 110
3 ADo 192.168.20.0/24     10.10.30.1 110
4 ADo 192.168.30.0/24     10.10.30.1 110
5 ADo 192.168.40.0/24     10.10.30.1 110
6 ADo 192.168.50.0/24     10.10.20.1 110
```

## 8. Verifikasi Koneksi end-to-end

Lakukan pengujian dari pc vlan menuju lan hauwei.

Dari pc vlan 10:

ping 192.168.50.10

Dari pc vlan 20:

ping 192.168.50.10

Dari pc vlan 30:

ping 192.168.50.10

Dari pc vlan 40:

ping 192.168.50.10

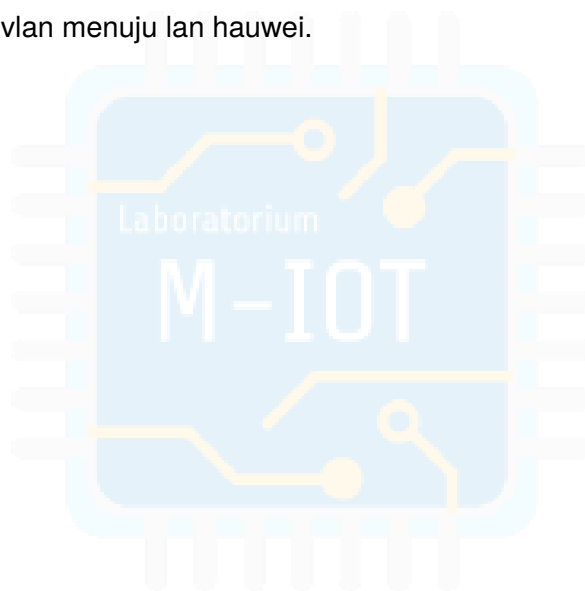
Lakukan pengujian dari pc lan huawei ke ip vlan cisco:

ping 192.168.10.11

ping 192.168.20.11

ping 192.168.30.10

ping 192.168.40.10



## Praktikum 5 Pengujian Failover OSPF

### 1. Kondisi normal: Dua link cisco-huawei aktif

Cek route dari cisco menuju lan huawei:

show ip route 192.168.50.0

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

```
Router>show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:03:45 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:03:45 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:03:45 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router>trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.10.2 8 msec
    10.10.11.2 4 msec
    10.10.10.2 3 msec
  2 192.168.50.10 9 msec 5 msec 5 msec
```

## 2. Kondisi 1 link cisco-huawei mati

Matikan salah satu link cisco-huawei.

```
configure terminal
interface gi0/3
shutdown
exit
```

Cek route dari cisco menuju lan huawei:

```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# shutdown
Router(config-if)#exit
Router(config)#
*Jun 11 01:07:07.100: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from FULL to DOWN, Neighbor Down: Interface down or detached
Router(config)#
*Jun 11 01:07:09.074: %LINK-5-CHANGED: Interface GigabitEthernet0/3, changed state to administratively down
*Jun 11 01:07:10.074: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to down
Router(config)#exit
Router#
*Jun 11 01:07:15.742: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:07:13 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:07:13 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.11.2 4 msec 2 msec 7 msec
  2 192.168.50.10 8 msec 4 msec 5 msec
```

### 3. Kondisi dua link cisco-huawei mati

Matikan link cisco-huawei yang kedua:

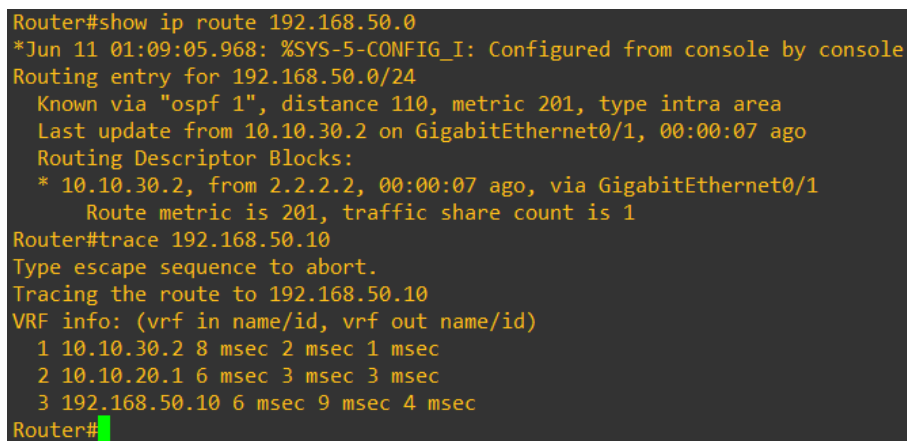
```
configure terminal
interface gi0/0
shutdown
exit
```

Cek route dari cisco menuju lan huawei:

```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```



```
Router#show ip route 192.168.50.0
*Jun 11 01:09:05.968: %SYS-5-CONFIG_I: Configured from console by console
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:07 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:07 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  1 10.10.30.2 8 msec 2 msec 1 msec
  2 10.10.20.1 6 msec 3 msec 3 msec
  3 192.168.50.10 6 msec 9 msec 4 msec
Router#
```

### 4. Mengaktifkan kembali dua link cisco huawei

```
configure terminal
interface gi0/3
no shutdown
exit
```

```
interface gi0/0
no shutdown
exit
```

Cek route dari cisco menuju lan huawei:

```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
*Jun 11 01:11:43.231: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 11 01:11:43.433: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 11 01:11:44.231: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to up
*Jun 11 01:11:44.433: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#show ip route 192.168.50.0
*Jun 11 01:11:50.460: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
*Jun 11 01:11:51.452: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:02:58 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:02:58 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#
*Jun 11 01:11:57.475: %SYS-5-CONFIG_I: Configured from console by console
Router#trace 192.168.50.10
Type escape sequence to abort.
Tracing the route to 192.168.50.10
VRF info: (vrf in name/id, vrf out name/id)
  0 10.10.10.2 6 msec
    10.10.11.2 3 msec
    10.10.10.2 3 msec
  2 192.168.50.10 11 msec 6 msec 4 msec
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:02:13 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:02:13 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:02:13 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1

```

## 2 Analisis Hasil Percobaan

Praktikum modul ini membahas integrasi jaringan multivendor yang terdiri dari switch dan router Cisco, router Huawei, serta router MikroTik. Fokus pembahasan meliputi konfigurasi VLAN beserta trunking, penggunaan router sebagai gateway tiap VLAN sekaligus server DHCP, pembuatan koneksi antarrouter, routing dinamis OSPF lintas vendor dengan pengaturan cost untuk membedakan jalur utama dan jalur cadangan, serta pengujian failover ketika jalur utama terputus.

Praktikum 1 dilakukan dengan membuat empat VLAN pada switch Cisco, mengatur port menuju PC sebagai access port sesuai VLAN masing-masing, dan mengkonfigurasi port menuju router sebagai trunk yang membawa seluruh VLAN. Praktikum 2 melanjutkan dengan membuat subinterface pada router Cisco sebagai gateway tiap VLAN, mengaktifkan DHCP server untuk dua VLAN, dan memberi IP statis pada dua VLAN lainnya. Praktikum 3 berfokus pada pemberian alamat untuk jaringan lokal di sisi Huawei serta seluruh link yang menghubungkan ketiga router. Praktikum 4 mengkonfigurasi OSPF pada ketiga router dengan pengaturan cost, di mana dua link Cisco-Huawei dijadikan jalur utama dan link melalui MikroTik dijadikan jalur cadangan. Praktikum 5 menguji perilaku failover dengan memutus link jalur utama secara bertahap dan mengamati perubahan jalur trafik menuju jaringan lokal Huawei.

Secara umum hasil yang diperoleh sudah sesuai dengan teori. VLAN dan trunking berjalan dengan benar, terbukti dari PC pada VLAN yang sama dapat saling terhubung melalui trunk, sementara PC dengan IP statis pun dapat menjangkau gateway dan PC lain antar VLAN melalui router. DHCP server pada router berhasil memberikan alamat kepada PC pada VLAN yang dikonfigurasi secara otomatis. Koneksi antarrouter juga aktif sepenuhnya, sehingga OSPF dapat membentuk hubungan tetangga penuh pada ketiga router. Pengaturan cost terbukti berhasil mengarahkan trafik melalui jalur dengan metric terendah, yaitu jalur langsung Cisco-Huawei, dan baru beralih ke jalur melalui MikroTik

saat kedua link utama terputus, kemudian kembali ke jalur utama setelah link diaktifkan ulang.

### 3 Hasil Tugas Modul

<https://github.com/hideaki7/Praktikum-Jarkom-2026-Kelompok-4/blob/main/MODUL%205/README.md>

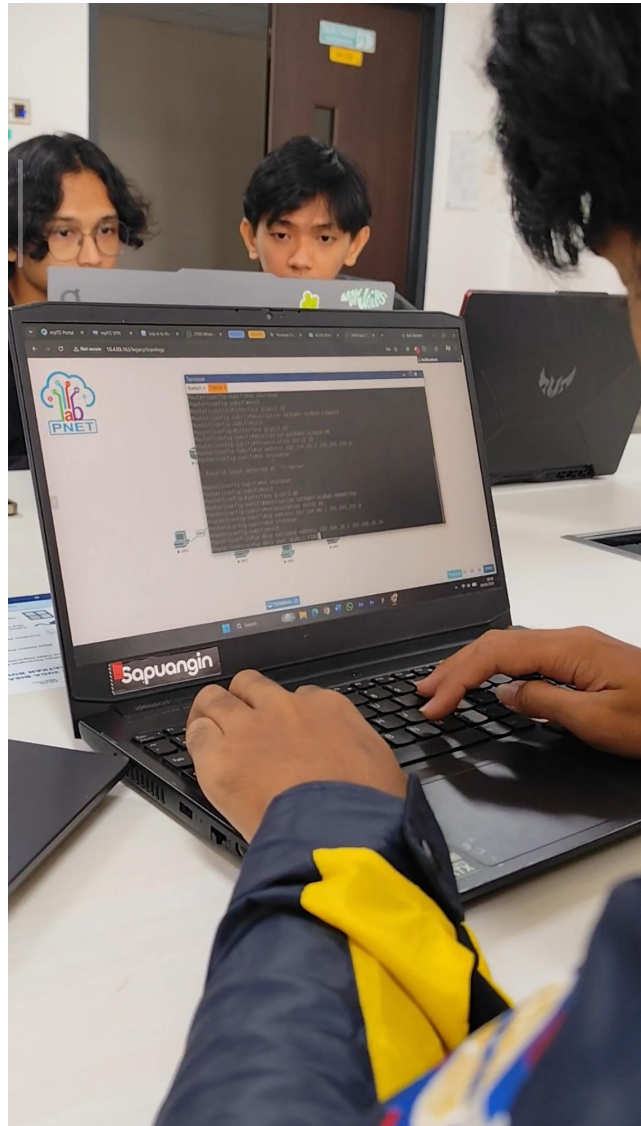
### 4 Kesimpulan

Praktikum modul ini berhasil diselesaikan dengan hasil yang sesuai dengan teori. Konfigurasi VLAN, trunking, dan router sebagai gateway tiap VLAN terbukti mampu menghubungkan banyak segmen jaringan melalui satu link fisik dengan gateway berbeda untuk setiap segmen. Integrasi tiga vendor router yang berbeda dapat saling terhubung dan menjalankan routing dinamis OSPF secara bersamaan, sementara pengaturan cost berhasil menentukan jalur utama dan jalur cadangan sesuai rencana. Pengujian failover menunjukkan bahwa OSPF dapat memindahkan trafik secara otomatis ke jalur cadangan ketika jalur utama terputus, dan mengembalikannya ke jalur utama setelah jalur tersebut aktif kembali, sehingga konsep redundansi jaringan menggunakan routing dinamis terbukti berjalan dengan baik.



## 5 Lampiran

### 5.1 Dokumentasi saat praktikum



**Gambar 1:** Dokumentasi